

Abstract

Seasonal heat storage solution for Domestic Hot Water systems

The following thesis is a comprehensive assessment of the technical, economic, and environmental feasibility of using sand batteries for large-scale heat storage in district heating networks, for the purpose of supplying domestic hot water in the United Kingdom. The research methodology used aims to be applicable to similar projects in the UK, providing a blueprint for their evaluation.

In the thesis, existing literature was explored on the various ways of producing domestic hot water, as well as existing cases where thermal energy storage was used in district heating. The English town of Guildford was selected as a case study for a proposed network and heat storage system. For this town, local and national-level energy use data was gathered and analysed from existing literature, and an EPANET simulation of the network was run. The data obtained was used to define the required performance of the proposed thermal energy storage system. After considering various heat storage systems, a sand battery was picked for basic sizing and performance calculations. Finally, the economic viability compared to other heat storage systems was analysed, as well as the system's environmental footprint.

The thesis finds that the sand battery system is technically plausible and would have minimal environmental impact, but its economic feasibility strongly depends on specifics related to the availability and effective use of charging opportunities, energy use strategies, and proximity to nearby waste heat sources.